

DESIGN AND IMPLEMENTATION

EDUCATIONAL ELECTRICITY LOAD FORCASTING SYSTEM

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Outline

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5. Machine Learning Model (Random Forest)
6. System Demonstration & Results
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Context and Motivation

Electricity demand constantly changes depending on the time of day, the weather, and human habits. If companies don't predict it correctly, they can over- or under-generate, causing losses or blackouts.

In 2012, a global competition called GEFCom2012 was held, where researchers and companies attempted to predict actual electricity demand using weather and consumption data.

- Accurate forecasting is essential for system stability, preventing overloads and minimizing operational costs.
- Industrial-grade forecasting systems can be complex, but understanding the fundamentals is key in educational settings.
- This project provides an accessible, end-to-end forecasting workflow that enables students to explore real ML concepts interactively.

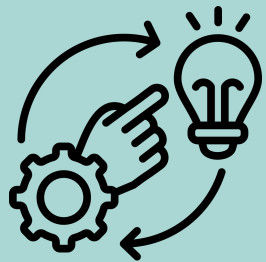


Project objective



| Interactivity

A GUI-based application (Tkinter) that eliminates command-line complexity.



| Simplicity



| Transparency

Direct visualization of real vs predicted demand curves.



| Educational Value

Shows how temporal and temperature-related features influence load behavior.

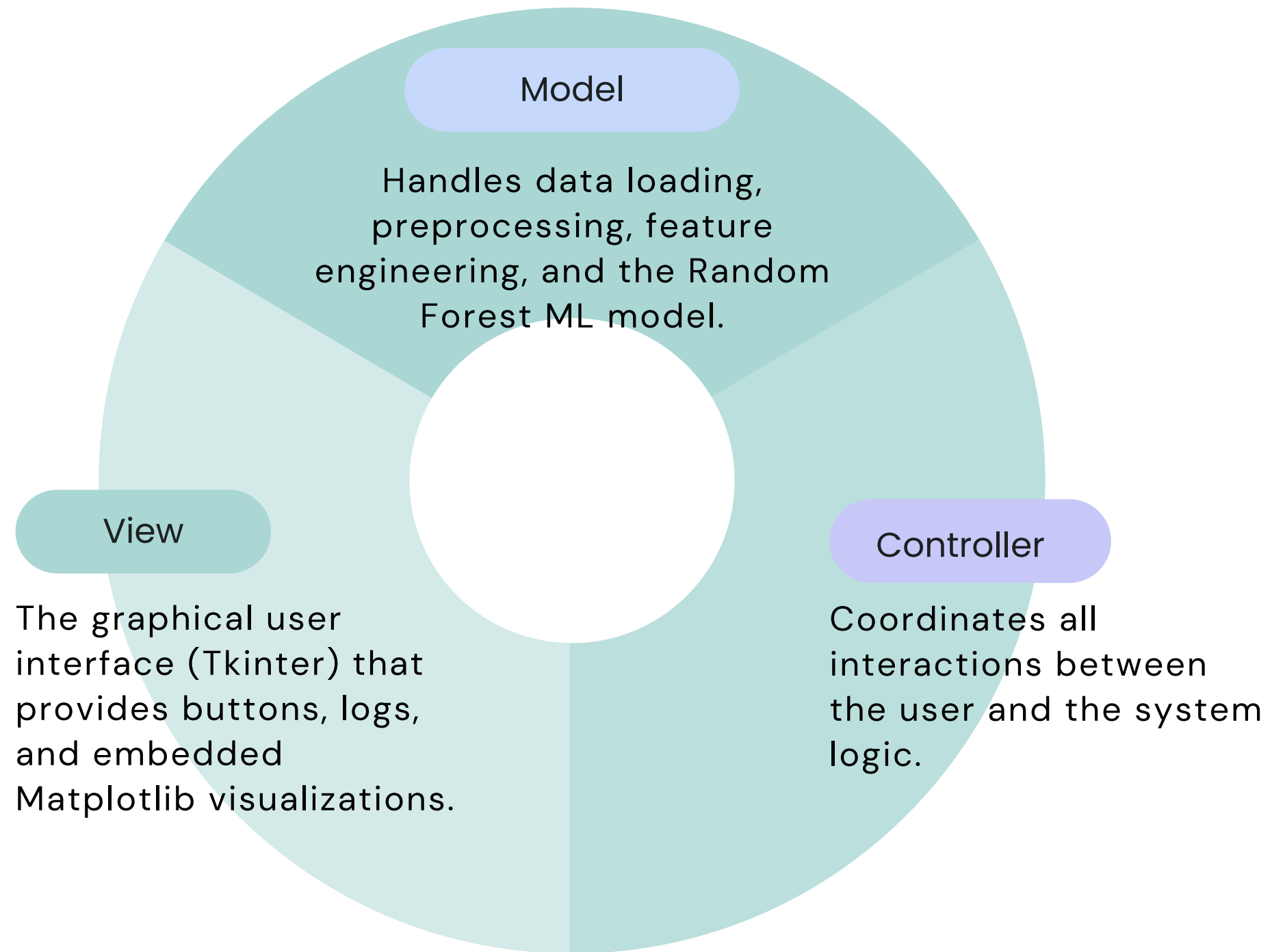


System Architecture (MVC)

```
electricity_demand_project/  
├── data/  
│   ├── raw/           # Original datasets  
│   └── output/        # Results (submission.csv)  
├── models/            # Data models and ML  
├── views/             # Graphical user interface (GUI)  
├── controllers/       # Control logic  
├── utils/             # Utilities  
├── tests/             # Test results  
├── docs/              # Documentation  
└── main.py            # Entry point
```



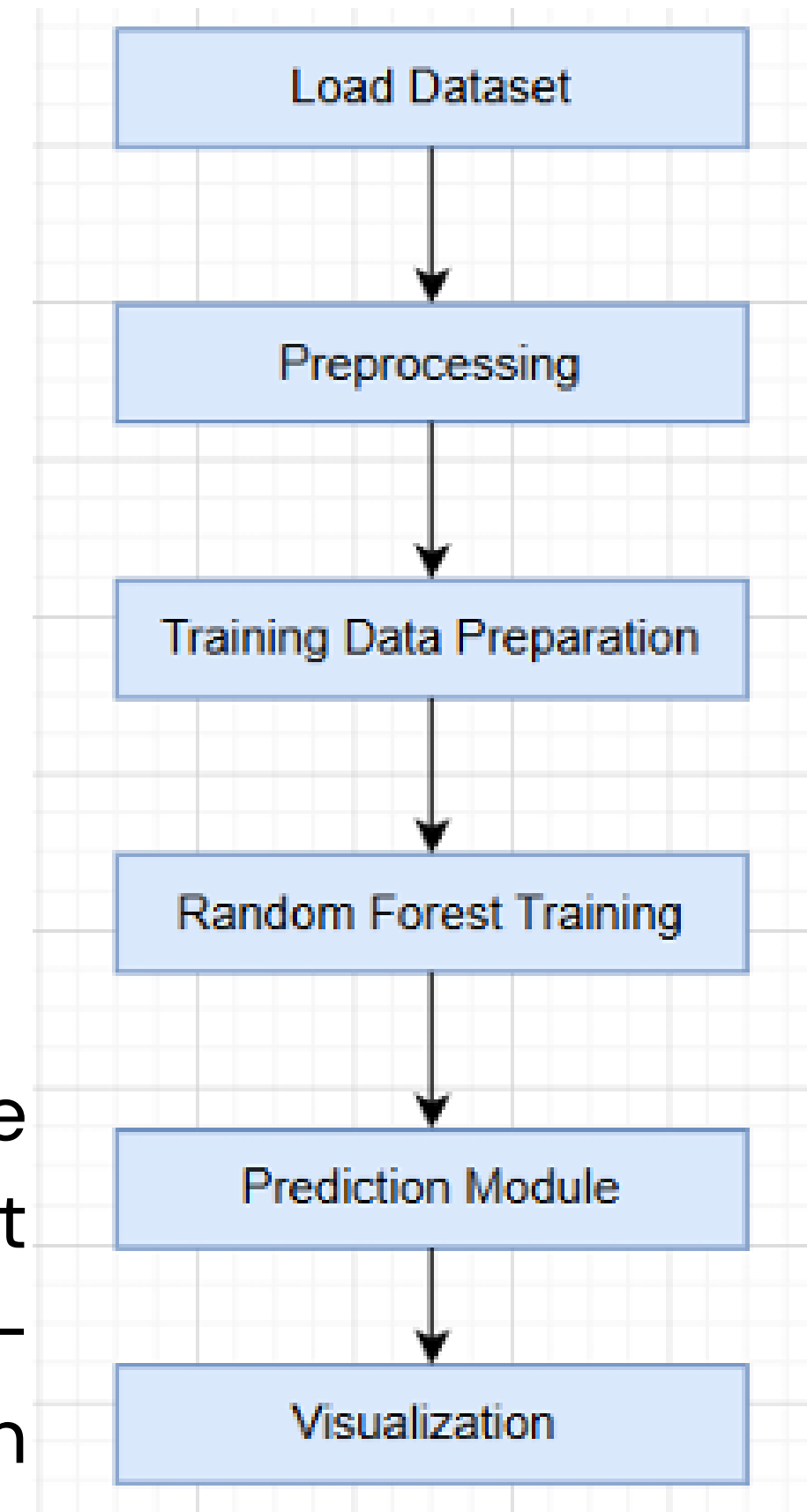
System Architecture (MVC)



Data Processing Pipeline

- Consistency.
- Reproducibility.
- Modularity.
- Scalability.
- Educational Clarity.

Overall, the pipeline ensures that the forecasting system is not only functional but also robust, transparent, and easy to extend—qualities that are especially important in an educational context.



Tech Stack

Tool / Library	Purpose	Why Used
Python	Core programming language	Easy to learn, widely used in ML and education
pandas	Data loading & preprocessing	Simple handling of CSV files and time-based features
NumPy	Numerical operations	Efficient array manipulation for feature engineering
scikit-learn	Machine Learning (Random Forest)	Robust model with minimal tuning; ideal for tabular data
Tkinter	Graphical User Interface (GUI)	Lightweight, built into Python, perfect for educational apps
Matplotlib	Plotting real vs predicted values	Enables clear visualization inside the GUI



MACHINE LEARNING MODEL (RANDOM FOREST)

REASONS FOR SELECTION

- Captures nonlinear relationships in load data.
- Minimal hyperparameter tuning required.
- Works well with engineered temporal features.

```
1 + from sklearn.ensemble import RandomForestRegressor
2 +
3 + class MLModel:
4 +     def __init__(self):
5 +         self.model = RandomForestRegressor(n_estimators=100, random_state=42)
6 +
7 +     def train(self, X_train, y_train):
8 +         self.model.fit(X_train, y_train)
9 +         return self
10 +
11 +     def predict(self, X_test):
12 +         return self.model.predict(X_test)
```

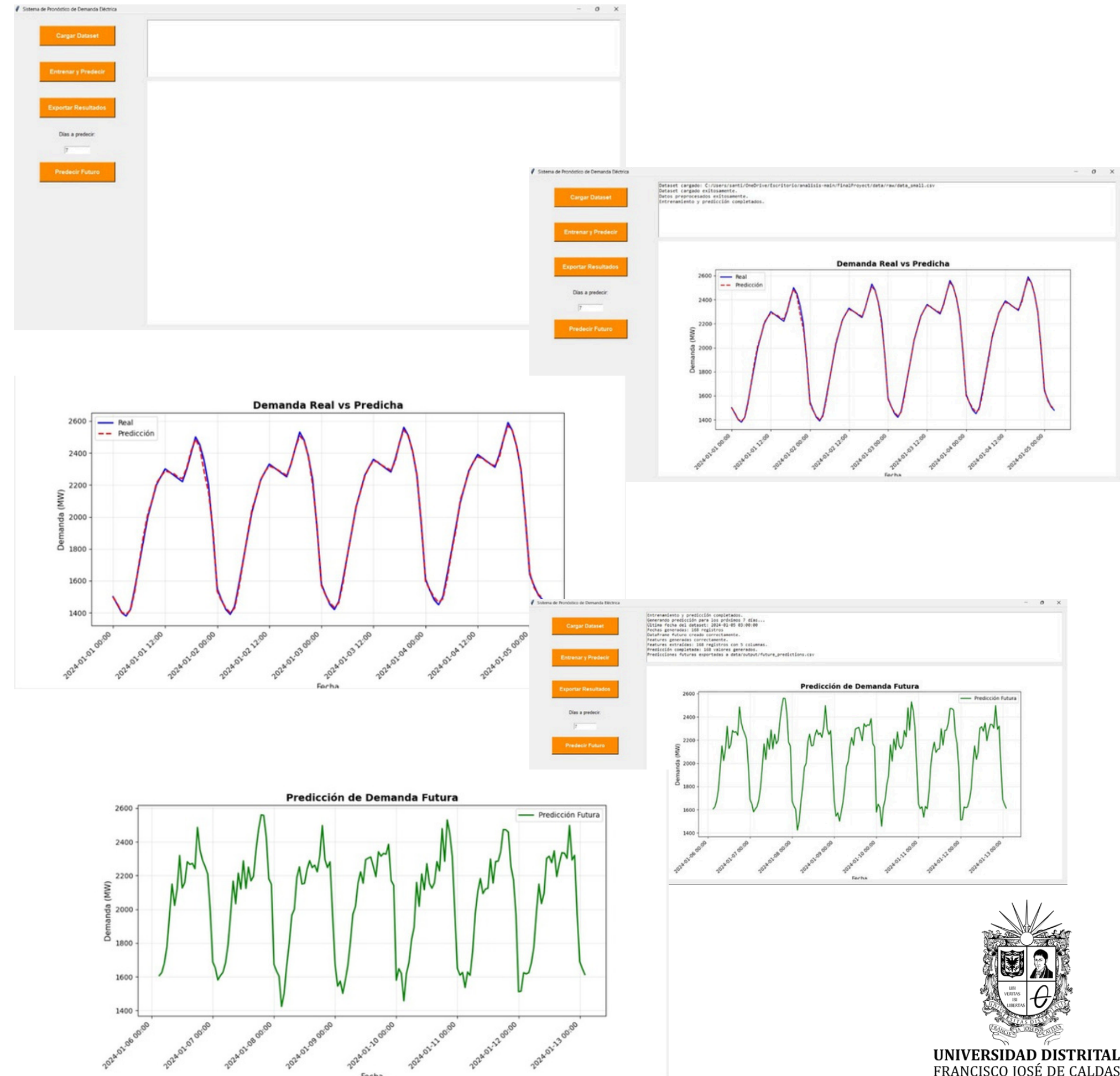
CONFIGURATION

- n_estimators = 100
- Input features: hour, day of week, day of year, weekend flag, normalized temperature.



Mockups

This mockup is made to show what the user experience (UX) is like for stakeholders.



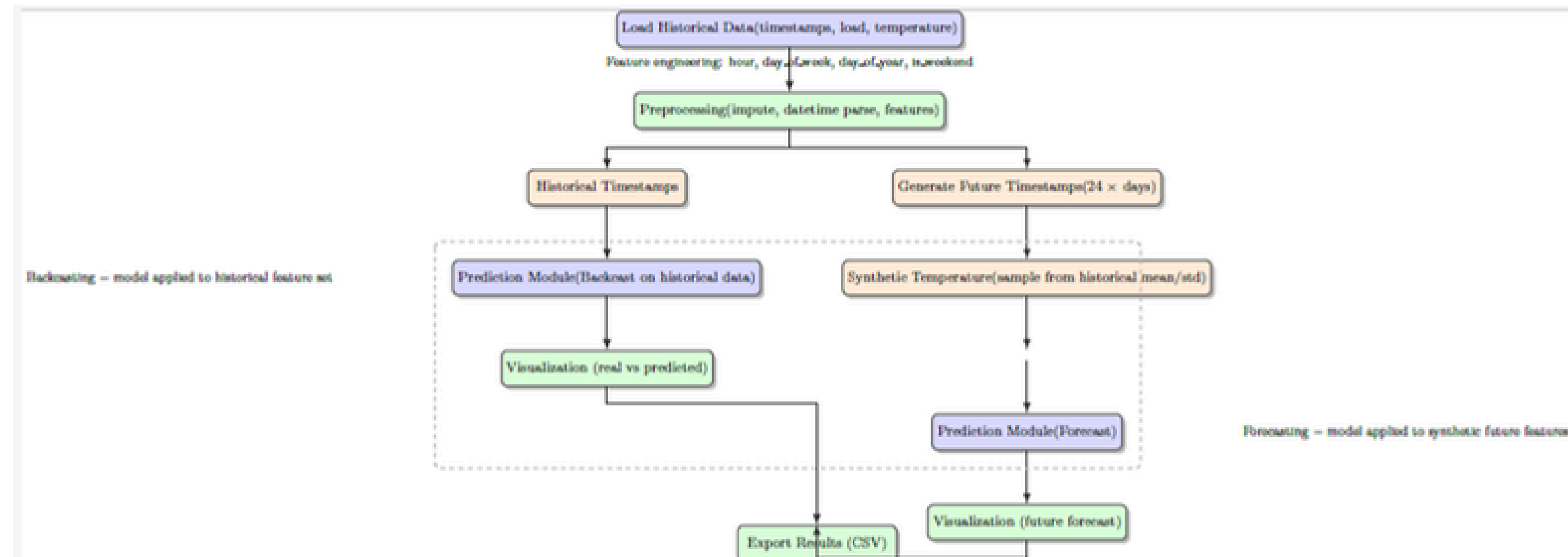
Forecast Generation

1. Backcasting (Historical Reconstruction):

- Validates model learning by comparing predictions against historical data.
- Display: real vs predicted curves.

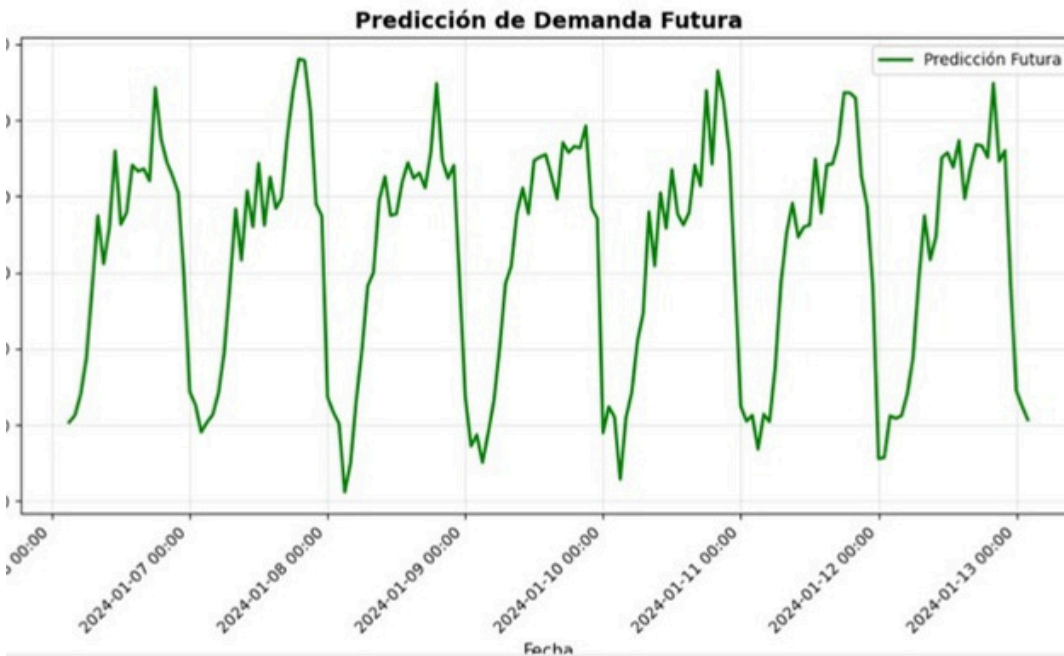
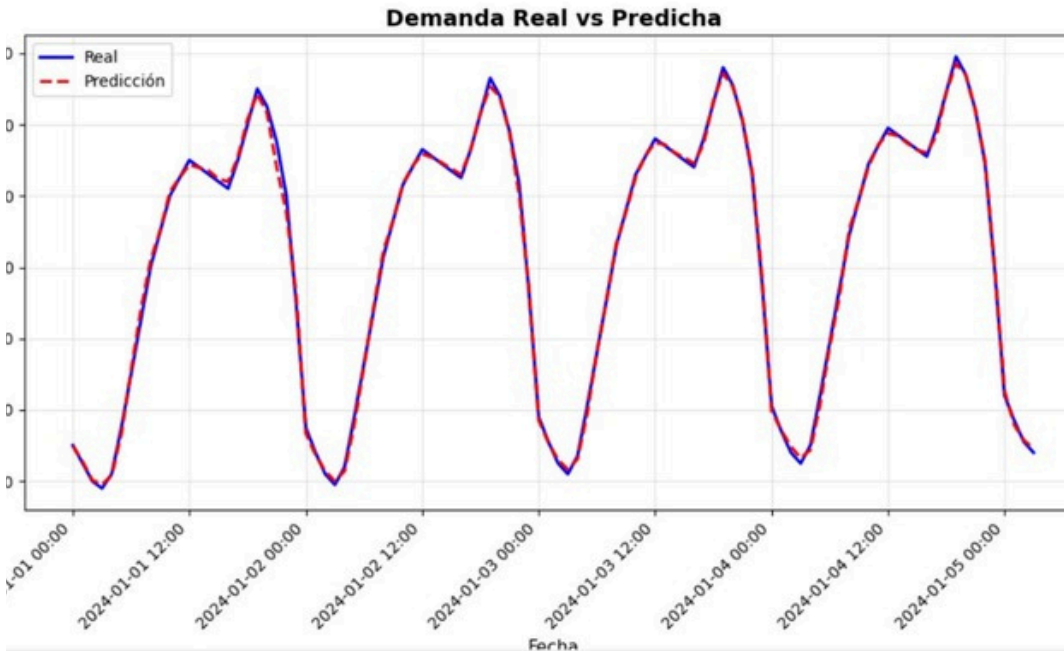
2. Forecasting (Future Prediction):

- User selects number of future days.
- Synthetic temperature values are generated using the mean and standard deviation of the dataset.
- Produces 24 × days hourly predictions.



Results & Evaluation

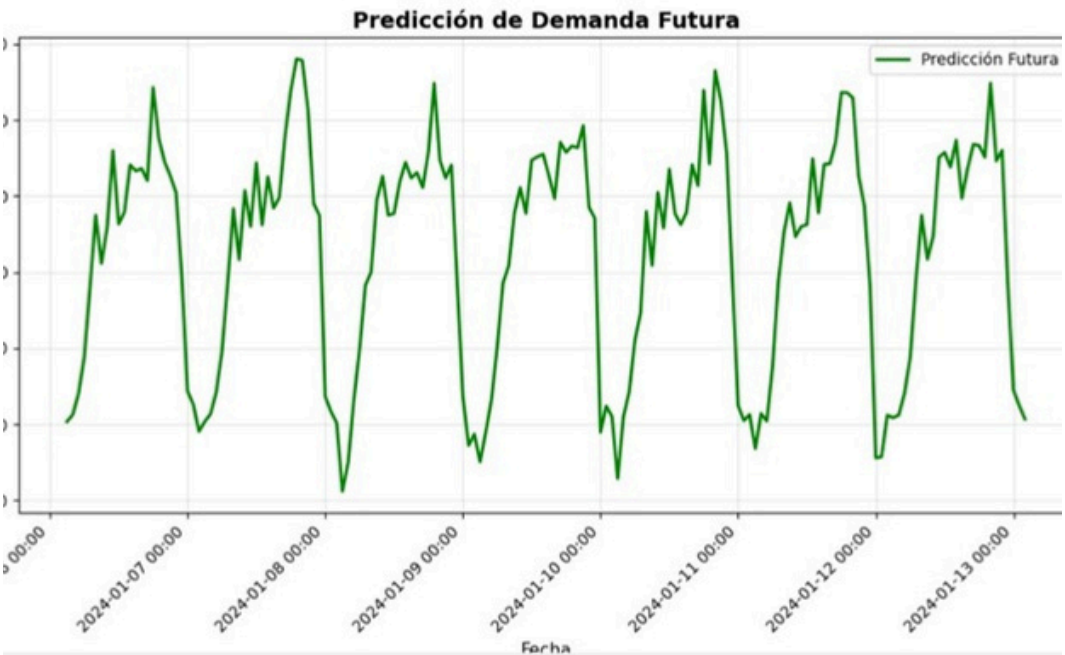
REAL VS PREDICTED



EXPORT CSV

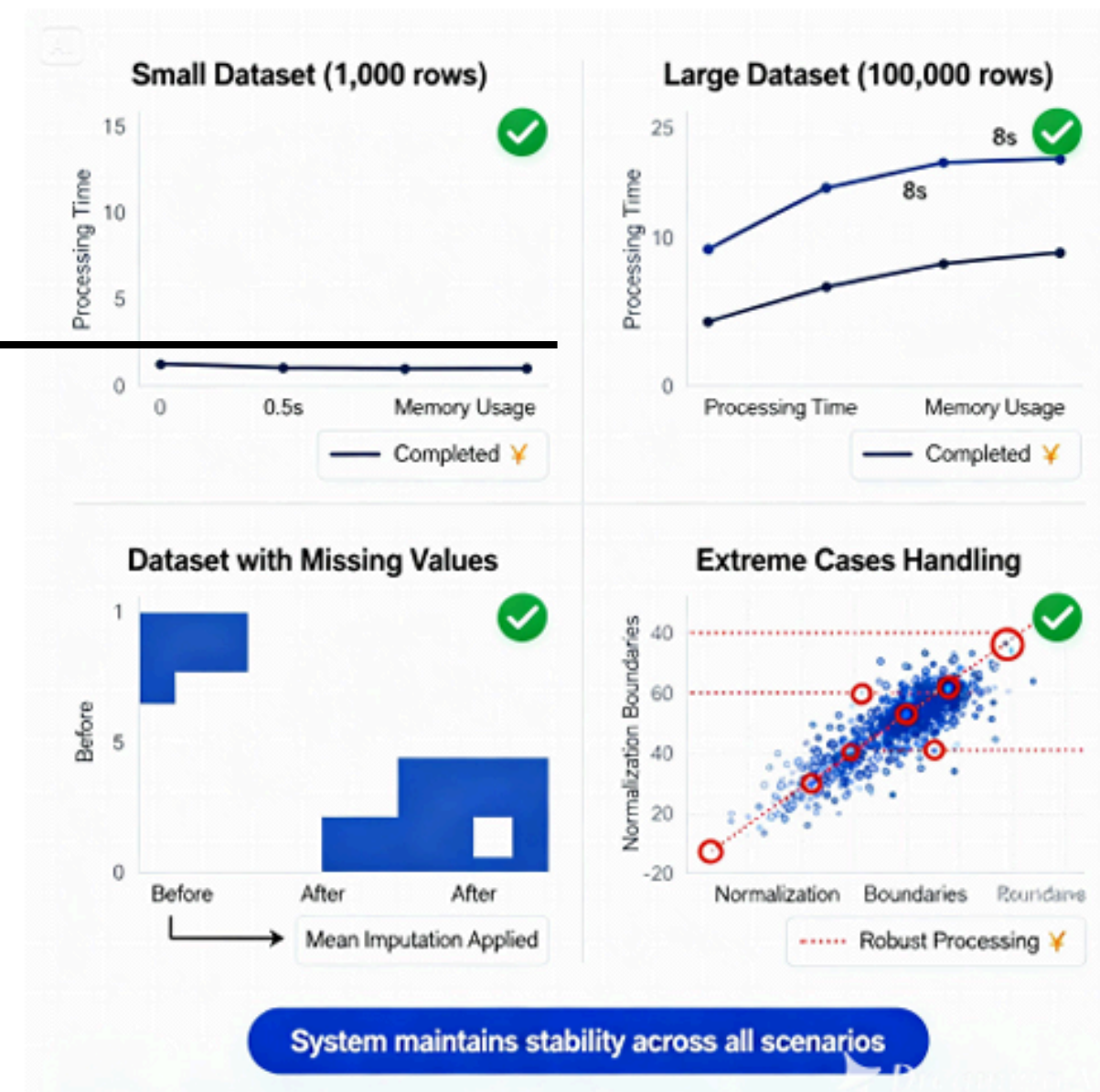
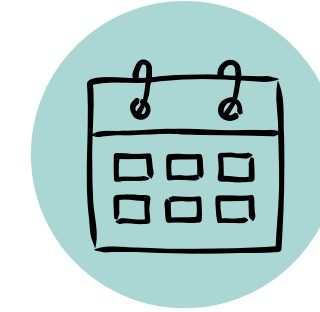
```
data_extreme.csv submission.csv X data_large.csv
data > output > submission.csv > data
1 datetime,forecast
2 2024-01-01 00:00:00,1498.3450000000003
3 2024-01-01 01:00:00,1455.4960000000005
4 2024-01-01 02:00:00,1403.0980000000022
5 2024-01-01 03:00:00,1390.9810000000004
6 2024-01-01 04:00:00,1416.003
7 2024-01-01 05:00:00,1533.21
8 2024-01-01 06:00:00,1710.7480000000014
9 2024-01-01 07:00:00,1879.677
10 2024-01-01 08:00:00,2018.5439999999976
11 2024-01-01 09:00:00,2103.5480000000025
12 2024-01-01 10:00:00,2211.6849999999997
13 2024-01-01 11:00:00,2251.2459999999998
14 2024-01-01 12:00:00,2286.7560000000017
15 2024-01-01 13:00:00,2279.2220000000007
16 2024-01-01 14:00:00,2270.5309999999998
17 2024-01-01 15:00:00,2247.4800000000027
18 2024-01-01 16:00:00,2239.8220000000003
19 2024-01-01 17:00:00,2308.8179999999998
20 2024-01-01 18:00:00,2414.3829999999957
21 2024-01-01 19:00:00,2485.6860000000033
22 2024-01-01 20:00:00,2433.1129999999999
23 2024-01-01 21:00:00,2281.1010000000015
```

FUTURE FORECAST



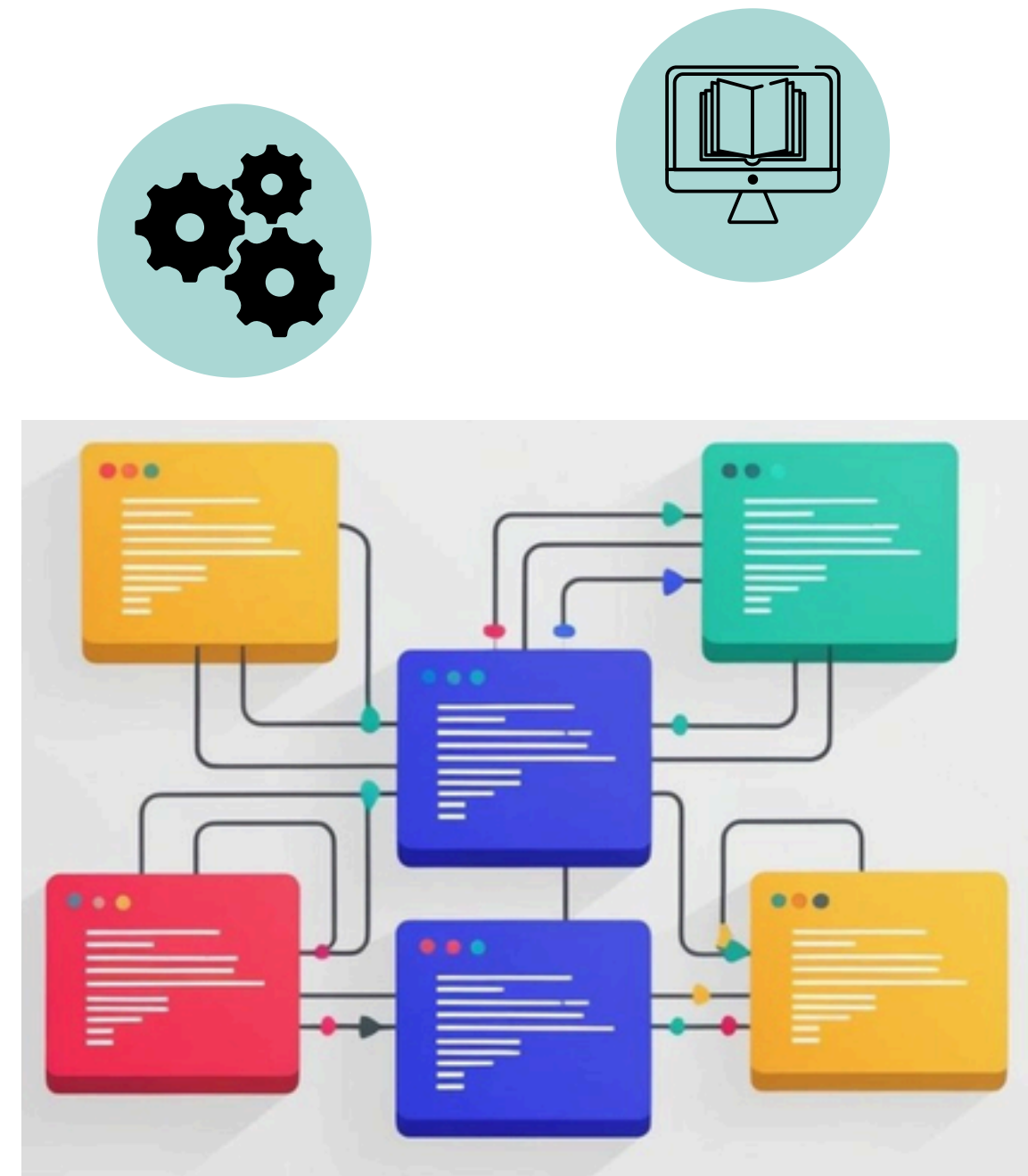
Results

- The model successfully reproduces the main temporal patterns in historical demand.
- Visual evaluation confirms coherence between predicted and real curves.
- Integration testing validated the full workflow: loading, preprocessing, training, forecasting, exporting.
- While the system does not compute RMSE or MAPE, it fulfills its educational purpose through clear visualization.



Conclusions and Q&A

- The MVC architecture provided clear separation between interface, logic, and machine learning components.
- The GUI enables a no-code approach to forecasting, suitable for students and beginners.
- The Random Forest model demonstrated stable performance for educational purposes.
- The system integrates all course workshops into a coherent end-to-end forecasting application.
- Future improvements include:
 - Adding evaluation metrics (RMSE, MAE)
 - Exploring additional ML models
 - Extending the system into a web-based environment



**Thank you
very much!**